I have uploaded two CSV files:

1. **Generated captions:** Contains image IDs and AI-generated medical captions.
2. **Ground-truth captions:** Contains the corresponding ROCOv2 image IDs and reference captions.

I want you to perform a systematic **Hallucination Assessment** by comparing the generated captions against the ground-truth captions.

Instructions

1. Match generated captions with ground-truth captions using their image IDs. Where multiple reference captions exist, consider all references associated with that image.
2. Examine the clinical meaning of each caption, identifying distinct medical concepts, including anatomical structures, clinical findings, abnormalities, anatomical locations, laterality and relevant diagnostic details.
3. Recognise medical synonyms and clinically equivalent descriptions rather than relying on exact word matching.
4. Account for negation and contradictory findings. Do not treat an affirmative finding and its negation as equivalent.
5. Identify **unsupported concepts**: medical concepts asserted in generated captions that are not supported by the corresponding ground-truth captions, including contradictory clinical findings.
6. Identify **omitted concepts**: medical concepts present in the ground-truth captions but absent from the generated captions.
7. Count each distinct medical concept only once per image. Do not count ordinary connecting words or non-clinical language.
8. Calculate the following metrics:

Unsupported Concept Rate (UCR) = Total unsupported generated concepts / Total generated clinical concepts × 100.

Concept Omission Rate (COR) = Total omitted ground-truth concepts / Total ground-truth clinical concepts × 100.

9. Aggregate the counts across all evaluated images before calculating the overall percentages. Do not simply average individual image percentages.
10. Distinguish genuinely contradictory information from information that is merely absent from the reference. Report ambiguous cases separately rather than assuming they are hallucinations.

Required output

Provide a simple summary table with these columns:

Metric	Number of Concepts Affected	Total Concepts Evaluated	Rate (%)
Unsupported Concept Rate			
Concept Omission Rate			

Also report the number of images evaluated and briefly explain any exclusions or uncertain cases.

Important requirements

- Evaluate all matched images rather than using a sample.
- Do not invent or estimate results.

- Do not assume that the ground-truth captions describe every finding visible in the original medical images.
- Treat these metrics as **reference-based measures of unsupported concepts and omissions**, not definitive proof of clinical hallucinations.
- If the files cannot be fully processed, explain the limitation and identify how many images were actually evaluated.
- Keep the final response concise, factual and suitable for inclusion in an academic project report.